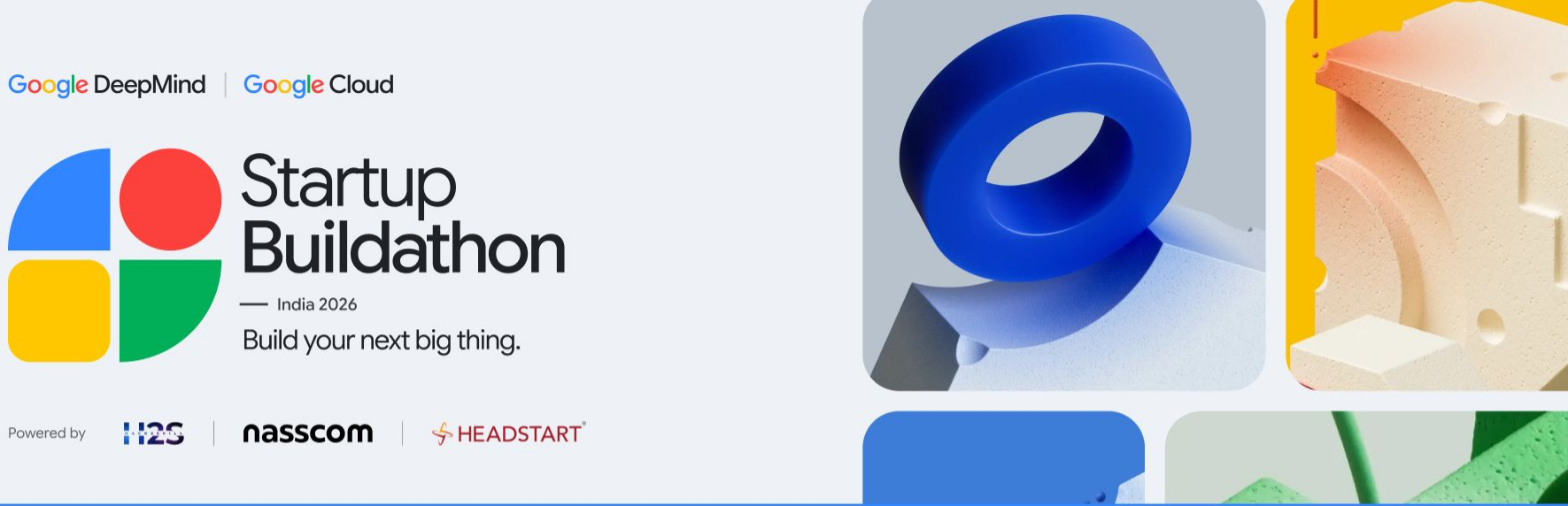


Team Name:	**AI Travel Planner**
Team Size:	1 (Solo founder: Shaik Irfan · shaikirfan24bcs0136@iiitkottayam.ac.in)
Problem Statement:	Planning one trip in India means juggling separate apps for trains, flights, buses, hotels and the day-by-day plan, with no single place to book it all and get a personal itinerary.

Problem Statement

The challenge or problem identified by the team.

Fragmented booking

Trains on IRCTC-based apps, flights on another site, buses on a third, hotels on a fourth. Travellers compare across tabs and re-enter the same passenger details every time.

Scattered confirmations

PNRs, e-tickets and hotel vouchers arrive in different emails and SMS, so finding the right ticket at the station or check-in desk is stressful.

Planning is a separate job

After booking, travellers still spend hours on blogs and videos to plan each day. Generic plans ignore their arrival time, hotel location, budget and food preferences.

Group trips multiply the pain

Families and friend groups must coordinate seats, rooms and details for everyone, which multiplies the effort and the chance of costly mistakes.

Who feels it most: India's domestic travellers: students and young professionals on budget trips, families on vacations and pilgrimages, friend groups, and first-time visitors to a city who want a ready, trustworthy plan.

Proposed Solution

The idea and how it addresses the identified problem.

AI Travel Planner takes a traveller from *"Delhi to Jaipur for 3 days, 2 people"* to booked tickets plus a personal plan, in one flow:

01
Search once
From, to, date, days, travellers, one-way or round trip

02
Compare & pick
Travel, Stay and Food tabs; book travel, stay or both

03
Details once
Traveller IDs, meal preference, special requests

04
Pay once
UPI or card for everything selected

05
Go
Tickets on email & WhatsApp + AI itinerary

🚆 **Live train data** LIVE
Real Indian Railways trains, seat availability (AVL / RAC / WL) and exact fares, cached to stay fast and cheap.

🏨 **Confident stay choice**
Hotels & hostels priced for the group, with photos, map, amenities, nearby sights and house rules.

✨ **Itinerary from your bookings**
Planned around arrival/departure times, hotel area, travellers' ages, diet and requests, with food picks and local tips.

Why it's different: booking apps stop at the ticket and AI chatbots can't see availability or book. We combine both: one checkout *and* a plan built from what you actually booked, all kept in one "My Trips" wallet.

Solution Architecture

Architecture of the proposed solution

Traveller

Web app on phone or laptop

Search · select · pay
My Trips wallet
PDF tickets

Email & WhatsApp confirmations

⇄

Next.js app (React + TypeScript) · Google Cloud Run

UI pages + server API routes in one autoscaling container

- Auth (JWT cookies, bcrypt)
- Search & inventory engine
- Live rail adapter + cache
- Pricing & checkout
- Booking & PNR service
- PDF ticket generator
- AI itinerary service
- Notification service

Prices are re-verified on the server at booking · AI output validated against a schema with a safe fallback · every third-party call has a fallback so the demo never breaks

⇄

- **Cloud SQL (PostgreSQL)** via Prisma: users, trips, bookings, API cache
- **Gemini API**: personalised itinerary (structured JSON)
- **IRCTC API (RapidAPI)**: trains, seat availability, fares
- **Razorpay**: UPI & cards (demo checkout today)
- **Resend** email · **Twilio** WhatsApp
- **Next:** flight, hotel & bus partner APIs

☁ Cloud Run (asia-south1) · Cloud SQL · Secret Manager (API keys) · Cloud Build CI from GitHub · Memorystore cache · Cloud Storage + CDN (tickets) · Cloud Tasks (async email/WhatsApp/AI)

Google AI Tools

Google AI Studio, Google Antigravity and Google Agent Skills (used or planned to be used).

Google AI Studio PLANNED

- Prototype and tune the itinerary prompt against real trip data
- Test Gemini's structured JSON output against our itinerary schema
- Compare Gemini models for speed, quality and cost
- Generate and manage the Gemini API key

Google Antigravity PLANNED

- Agent-assisted development of the next features: chat-based trip editing, real flight & hotel integrations
- Let agents run and verify end-to-end booking flows in the browser
- Speed up a solo founder's build and test cycle

Google Agent Skills PLANNED

- Package our capabilities as skills an AI travel agent can call: **search trains**, **check seats & fare**, **find stays**, **build itinerary**, **book trip**
- Lets a Gemini agent plan *and* book a trip from one natural-language request

Built so far: the full MVP was built solo in VS Code with Next.js, PostgreSQL and Prisma. The AI itinerary pipeline (context → prompt → schema-validated JSON → fallback) is already in place; moving it onto Gemini through AI Studio is the next step.

Google AI Models

Google AI model(s) used or planned for the proposed solution.

Gemini API (Flash) · core model

- **Itinerary generation:** the booked trip (legs, hotel, travellers, diet, requests, saved restaurants) in, a structured day-by-day plan out
- **Chat to edit the trip:** "make day 2 relaxed", "add a veg dinner near the fort"
- **Function calling:** Gemini calls our search / seat-check tools to suggest real, bookable options
- **Multilingual plans:** Hindi, Telugu, Tamil and more

Gemini multimodal · ticket import

- Read a photo, screenshot or PDF of a ticket booked elsewhere and add it to the trip wallet and itinerary
- Read hotel confirmation emails into a stay record

Nano Banana (Gemini image) · trip visuals

- Generate a shareable trip card / cover image for each itinerary
- Destination visuals when partner photos are missing

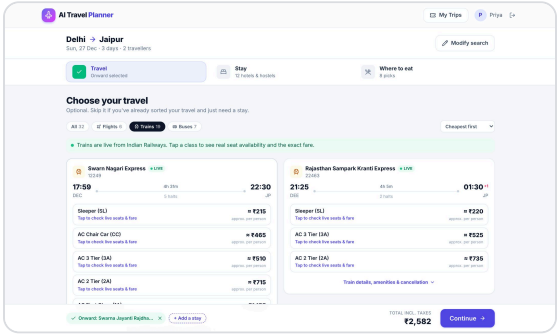
Why Gemini

- Long context fits the whole trip in one request
- Native JSON output plus schema validation
- Fast, low-cost Flash models suit a consumer app
- Strong local and Indian-language knowledge

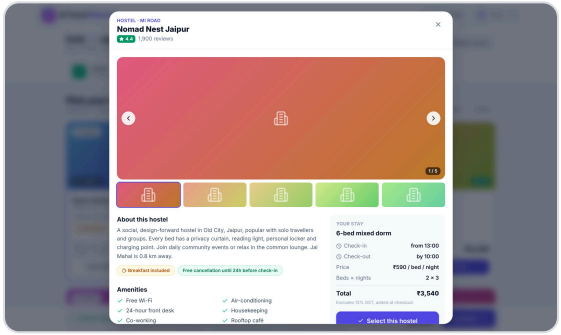
Status: the itinerary pipeline is built and running in the MVP. Gemini integration is in progress; the same prompt and schema switch over directly.

Working MVP

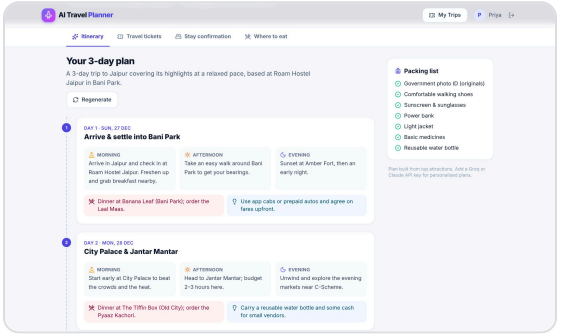
What is built and running today.



Live trains with real seats & fares



Stay details: gallery, map, rules



Day-by-day itinerary after booking

WORKING

Sign-up/login · full booking flow (travel only, stay only, or both) · live Indian Railways trains, seat availability & fares · AI itinerary · PDF e-tickets & stay vouchers · email & WhatsApp confirmations · My Trips wallet

DEMO DATA FOR NOW

Flights, buses, hotels & restaurants use realistic sample data · payment uses a demo checkout · no real tickets issued yet (needs IRCTC agent & travel-partner agreements)

Code: github.com/magicalnature621-coder/ai-travel-planner

Market, Business Model & Roadmap

How it grows into a venture.

Market

- **Primary:** India's domestic travellers: students, young professionals, families, groups
- **Secondary:** Nepal, Sri Lanka, Bhutan, UAE, and NRIs planning trips home

Revenue

- Booking commissions from hotel, flight & bus partners
- Convenience fee per journey (in checkout today)
- Premium AI planner subscription
- Affiliate: tours, cabs, insurance · B2B group trips

Roadmap

- Gemini itinerary + chat edits
- Real flight & hotel APIs; bus partners
- IRCTC / B2B partner for real ticketing
- Razorpay live · Cloud Run launch · mobile app

	MakeMyTrip / Goibibo	ixigo / ConfirmTkt	redBus	AI chatbots	AI Travel Planner
Travel + stay in one checkout	Separate flows	Train-first	Buses only	Can't book	✓
Itinerary built from your bookings	—	—	—	Generic	✓
Tickets + plan together on WhatsApp	Partly	Partly	Partly	—	✓

Google DeepMind | Google Cloud

Startup Buildathon

— India 2026

Build Your Next Big Thing.

Powered by H2S | nasscom | HEADSTART

Thank you

